

THE UNIVERSITY OF HONG KONG
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

Final Examination

COMP3314 Machine Learning

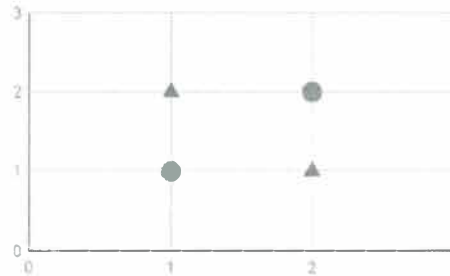
10 December 2020, 2:30 pm - 4:30 pm

- Candidates are permitted to refer to the following electronic/printed materials in the examination: textbook, lecture slides, assignment handout and sample solutions, and self-made notes
- Internet searching is NOT allowed
- All answers must be in your own words
- Write your University No. at the top of your answer script
- Use of a calculator is allowed. Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script
- Handwrite your answers on paper. Scan your handwritten answers into a PDF file for submission to OLEX

Question	Max. Mark
1	12
2	10
3	6
4	8
5	14
Total	50

Question 1:

1.1 (4 marks): Write down 5 classifiers discussed in class that are able to achieve 100% accuracy on the data below. Also, write down 3 classifiers discussed in class that are not able to achieve 100% accuracy on this data.



1.2 (2 marks for correct answer, -2 mark for incorrect answer): (True/ False)The goal of PCA is to interpret the underlying structure of the data in terms of the principal components that are best at predicting the output variable.

1.3 (2 marks for correct answer, -2 mark for incorrect answer): (True/ False)The output of PCA is a new representation of the data that is always of lower dimensionality than the original feature representation.

1.4 (2 marks): In a medical application domain, suppose we build a classifier for patient screening (True means patient has cancer). Which of the following situations would you like your classifier to have? Select all correct answers. (FP: false positive, FN: false negative, TP: true positive, TN: true negative)

- a) $FP \gg FN$
- b) $FN \gg FP$
- c) $FN = FP \times TP$
- d) $TN \gg FP$
- e) $FN \times TP \gg FP \times TN$
- f) All of the above

1.5 (2 marks): Suppose that there are a total of 50 machine learning related documents in a library of 200 documents. Now, suppose that a search engine retrieves 10 documents after a user enters 'machine learning' as a query, of which 5 are machine learning related documents. Write down the precision and recall for this example.

Question 2:

2.1 (2 marks): Say we have three models in an ensemble. We got the following results on a binary classification task.

Model Name	Class – 1 Probability	Class – 0 Probability
Model – 1	0.49	0.51
Model – 2	0.99	0.01
Model – 3	0.49	0.51

Apply soft voting and hard voting to this ensemble.

2.2 (2 marks): If AdaBoost underfits the training data, which hyperparameter should be changed and how?

2.3 (3 marks): For binary classification, which of the following statement(s) are true of AdaBoost with decision trees?

- a) It usually has lower bias than a single decision tree
- b) It can train multiple decision trees in parallel
- c) It is popular because it usually works well even before any hyperparameter tuning

2.4 (3 marks): For binary classification, which of the following statement(s) are true of AdaBoost?

- a) It can be applied to neural networks
- b) It uses the majority vote of learners to predict the class of a data point
- c) The metalearner provides not just a classification, but also an estimate of the posterior probability

Question 3:

3.1 (3 marks): Why would you want to use:

- a) Ridge Regression instead of plain Linear Regression?
- b) Lasso instead of Ridge Regression?
- c) Elastic Net instead of Lasso?

3.2 (3 marks): Consider the following prediabetic data.

Age (x)	Glucose Level (y)
43	99
21	65
25	79
42	75
57	87
59	81

Calculate the Pearson product-moment correlation coefficient for this data.

Question 4:

4.1 (3 marks): Consider a Multilayer Perceptron (MLP) with a number of hidden layers. Explain why backpropagation would fail if the weights of the hidden layer are not randomly initialized.

4.2 (3 marks): Write down all hyperparameters that you can adjust in a Multilayer Perceptron (MLP). Say your MLP underfits the data, how would you change those hyperparameters that you have listed?

4.3 (2 marks): If you are using dropout in your network, would it slow down training?

Question 5:

5.1 (2 marks): Consider the following two TensorFlow programs A and B. Write down the output for both programs and explain the difference between the two programs.

A:

```
1 import tensorflow as tf
2 x = tf.data.Dataset.from_tensor_slices(tf.random.uniform([5, 2]))
3 y = tf.data.Dataset.from_tensor_slices(tf.range(5))
4 ds_joint = tf.data.Dataset.zip((x, y))
5 ds = ds_joint.batch(4).repeat(count=2)
6 for i, (batch_x, batch_y) in enumerate(ds):
7     print(i, batch_x.shape, batch_y.numpy())
```

B:

```
1 import tensorflow as tf
2 x = tf.data.Dataset.from_tensor_slices(tf.random.uniform([5, 2]))
3 y = tf.data.Dataset.from_tensor_slices(tf.range(5))
4 ds_joint = tf.data.Dataset.zip((x, y))
5 ds = ds_joint.repeat(count=2).batch(4, True)
6 for i, (batch_x, batch_y) in enumerate(ds):
7     print(i, batch_x.shape, batch_y.numpy())
```

5.2 (2 marks): Consider the following TensorFlow program. What is the purpose of line 5?

```
1 import tensorflow as tf
2 x = tf.data.Dataset.from_tensor_slices(tf.random.uniform([5, 2]))
3 y = tf.data.Dataset.from_tensor_slices(tf.range(5))
4 ds_joint = tf.data.Dataset.zip((x, y))
5 ds_trans = ds_joint.map(lambda x, y: (x*2-1.0, y))
```

5.3 (3 marks): Consider the following TensorFlow/Keras program. How many input nodes, output nodes and total parameters are in the model?

```
1 import tensorflow as tf
2 model = tf.keras.Sequential([
3     tf.keras.layers.Dense(16, activation='sigmoid', name='fc1', input_shape=(12,)),
4     tf.keras.layers.Dense(32, activation='sigmoid', name='fc2'),
5     tf.keras.layers.Dense(9, name='fc3', activation='softmax')])
6 model.summary()
```

5.4 (3 marks): Consider the following TensorFlow/Keras program. For each layer of the CNN, write down the number of parameters.

```
1 import tensorflow as tf
2 model = tf.keras.Sequential([
3     tf.keras.layers.Conv2D(16, (3, 3), padding='same', activation='relu'),
4     tf.keras.layers.MaxPooling2D((2, 2)),
5     tf.keras.layers.Dropout(rate=0.5),
6     tf.keras.layers.Conv2D(32, (3, 3), padding='same', activation='relu'),
7     tf.keras.layers.MaxPooling2D((2, 2)),
8     tf.keras.layers.Dropout(rate=0.5),
9     tf.keras.layers.Conv2D(64, (3, 3), padding='same', activation='relu'),
10    tf.keras.layers.MaxPooling2D((2, 2)),
11    tf.keras.layers.Conv2D(128, (3, 3), padding='same', activation='relu'),
12 ])
13 model.build(input_shape=(None, 64, 64, 3))
```

5.5 (4 marks): Write down four things you could try if your GPU runs out of memory while training a CNN.

END OF PAPER